Solution

$$\frac{P}{Q} = \frac{R}{S}$$

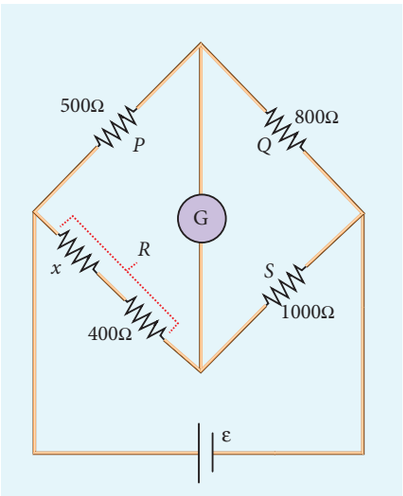
$$S = \frac{Q}{P} \times R$$

$$S = \frac{1000}{100} \times 40 \quad S = 400\ \Omega$$

EXAMPLE 2.24

What is the value of x when the Wheatstone's network is balanced?

$P = 500\ \Omega$, $Q = 800\ \Omega$, $R = x + 400$, $S = 1000\ \Omega$



Solution

$\frac{P}{Q} = \frac{R}{S}$, when the network is balanced

$$\frac{500}{800} = \frac{x + 400}{1000}$$

$$x + 400 = \frac{5}{8} \times 1000$$

$$x + 400 = 625$$

$$x = 625 - 400$$

$$x = 225\ \Omega$$

2.5.4 Meter bridge

The meter bridge is another form of Wheatstone's bridge. It consists of a uniform wire of manganin AB of one meter length. This wire is stretched along a metre scale on a wooden board between two copper strips C and D. Between these two copper strips another copper strip E is mounted to enclose two gaps G_1 and G_2 as shown in Figure 2.26.

An unknown resistance P is connected in G_1 and a standard resistance Q is connected in G_2. A jockey (conducting wire-contact maker) is connected to the terminal E on the central copper strip through a galvanometer (G) and a high resistance (HR). The exact position of jockey on the wire can be read on the scale. A Lechlanche cell and a key (K) are connected between the ends of the bridge wire.

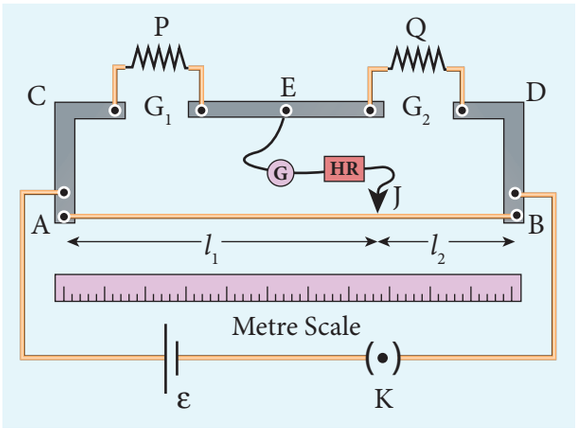


Figure 2.26 Meter bridge

The position of the jockey on the wire is adjusted so that the galvanometer shows zero deflection. Let the position of jockey at the wire be at J. The resistances corresponding to AJ and JB of the bridge wire form the resistances R and S of the Wheatstone's bridge. Then for the bridge balance

$$\frac{P}{Q} = \frac{R}{S} = \frac{r.\mathrm{AJ}}{r.\mathrm{JB}} \qquad (2.54)$$

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